WORKSHOP

DEVELOPER OPERATIONAL

TEKNIK INFORMATIKA GOLONGAN D

TERAFORM NGINX

PRESENTATION

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Bapak Aqshal Iksan

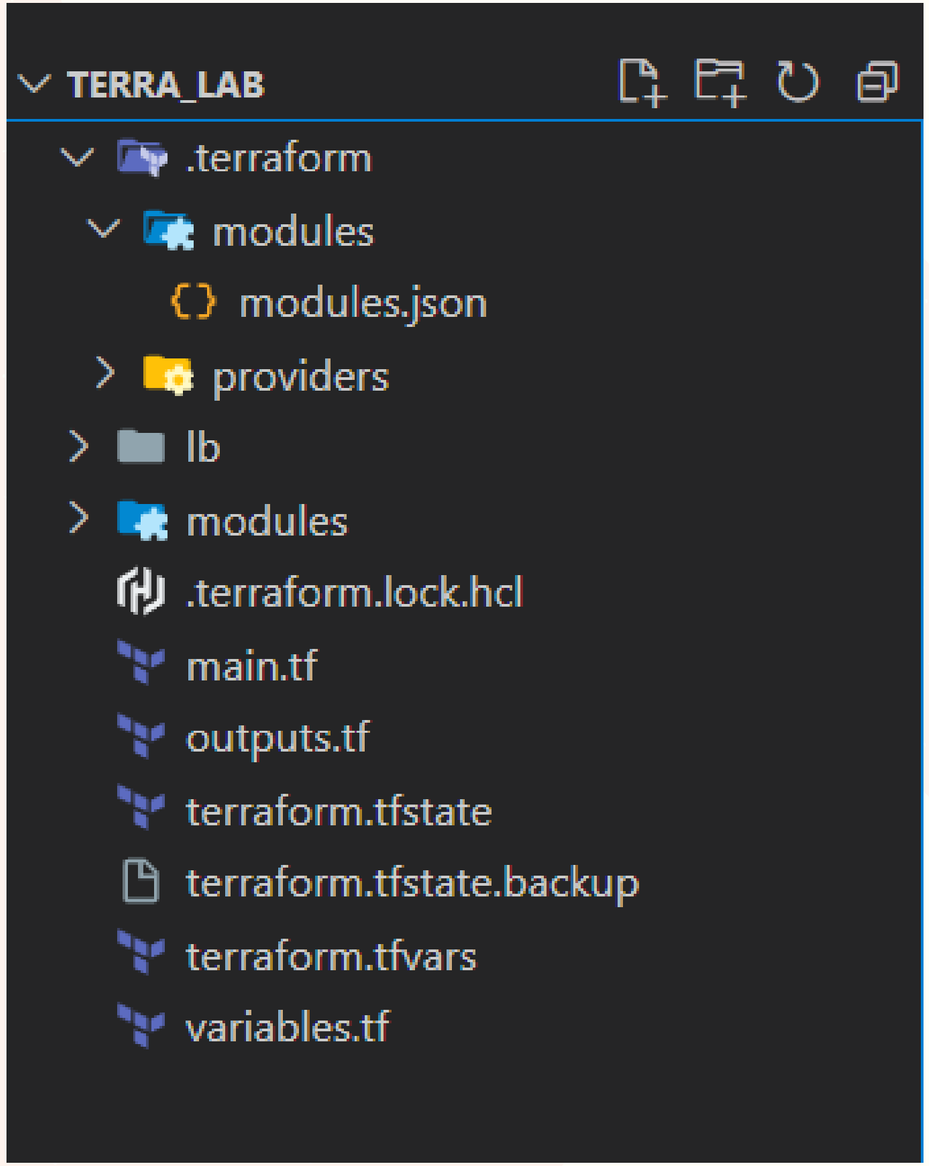
DEVELOPER OPERATIONAL

PRESENTATION

STRUKTUR PROJECT

Penjelasan:

- main.tf → File utama
- modules → Reusable components
- nginx.conf → Konfigurasi Load Balancer



TERAFORM INIT

```
[redacted]:~/Terraform-Nginx-Docker-LoadBalancer/terra_lab$ terraform init

Initializing the backend...
Initializing modules...

Initializing provider plugins...
- Reusing previous version of kreuzwerker/docker from the dependency lock file
- Using previously-installed kreuzwerker/docker v3.6.2

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

Fungsi:

- Mengunduh provider Docker
- Menyiapkan environment Terraform

TERRAFORM APPLY

```
:~/Terraform-Nginx-Docker-LoadBalancer/terra_lab$ terraform apply
module.loadbalancer.docker_image.nginx: Refreshing state... [id=sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d9988
5b3ae1734f3392enginx:latest]
docker_network.app_network: Refreshing state... [id=a69018812d1a78b3fb388f1612a323db86e288eee47ef78c8a0818ba3cce7be0]
module.backend.docker_image.nginx: Refreshing state... [id=sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae
1734f3392enginx:latest]
module.loadbalancer.docker_container.loadbalancer: Refreshing state... [id=b59fc99983d5011bcd4ed1c09f9a32e6d538331c69de7
4c351c33365e05d3c9e]
module.backend.docker_container.backend[1]: Refreshing state... [id=5102421af862c971107fdebd7520ae474077598e5e2df566119e
5199cb21668c]
module.backend.docker_container.backend[2]: Refreshing state... [id=f95cc8cec33f5437bf8dc51d97d13fd38a52df01c48d78943528
bebfcfe74724]
module.backend.docker_container.backend[4]: Refreshing state... [id=db9f91291a80730a2532a8176988e1c24707df5b9e8b6fc3f44c
fbd230bc687d]
module.backend.docker_container.backend[0]: Refreshing state... [id=ee71ba6b05b2f587f6fb9f925004734d1cf55dad833c86aeaef7
9550abbe24ab]
module.backend.docker_container.backend[3]: Refreshing state... [id=335bfc9a9478c21b94e785a0e08d11ee984fd9440811ba159028
4cb081dc1d2b]

Note: Objects have changed outside of Terraform

Terraform detected the following changes made outside of Terraform since the last "terraform apply" which may have
affected this plan:

  # docker_network.app_network has been deleted
  - resource "docker_network" "app_network" {
        id          = "a69018812d1a78b3fb388f1612a323db86e288eee47ef78c8a0818ba3cce7be0"
      - name        = "mini_network" -> null
```

Terraform akan:
Membuat Docker Network
Membuat 5 backend container
Membuat 1 load balancer container

DOCKER CONTAINER

```
container_names = [
nd_0",
nd_1",
nd_2",
nd_3",
nd_4",
```

Semua container berhasil dibuat.

nginx_lb	28ee41dc8332	341bf0f3ce6c	8080:80	0%	10 minutes ago		
backend_2	0343d58b11ae	341bf0f3ce6c		0%	10 minutes ago		
backend_3	e711523e1266	341bf0f3ce6c		0%	10 minutes ago		
backend_4	87ff77db096d	341bf0f3ce6c		0%	10 minutes ago		
backend_0	99fa9d21dd96	341bf0f3ce6c		0%	10 minutes ago		
backend_1	9d8b24c58bd9	341bf0f3ce6c		0%	10 minutes ago		

PRESENTATION

PENGUJIAN HASIL WEB

- Terraform mempermudah pembuatan infrastruktur
- Docker memudahkan pengelolaan container
- Nginx efektif sebagai Load Balancer
- Sistem berhasil membagi trafik ke beberapa backend

```
load_balancer_url = "http://localhost:8080"
```

THANK YOU

Implementasi ini membuktikan bahwa otomatisasi infrastruktur menggunakan Terraform serta containerisasi dengan Docker memberikan solusi yang scalable, efisien, dan mudah dikelola. Penggunaan Nginx sebagai load balancer memastikan distribusi trafik yang optimal dan meningkatkan reliabilitas sistem.

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